4. Results

4.1. Breakdown thresholds and threshold behavior

Table 2 summarizes the experimental results on thresholds for bubble formation and bright plasma luminescence and on the threshold sharpness for different pulse durations, wavelengths, and modes of laser operation. Figure 4(a) displays the pulse duration and wavelength dependence of bubble thresholds, and Fig. 4(b) illustrates the threshold behavior for slm and mlm laser operation.

λ (nm)	τ_L (FWHM)	Laser mode	NA	d (μm)	Breakdown criterion	E_{th} (nJ)	I_{th} (10^{11} W/cm^2)	S
1040	350 fs	ml	0.8	1.59	Bubble	25.0	80.0	30.5
520	306 fs	ml	0.8	0.79	Bubble	4.8	70.0	34.3
347	280 fs	ml	0.8	0.53	Bubble	3.6	131.5	66.4
355	0.56 ns	slm	0.3	1.44	Bubble	830	2.40	47.9
355	0.56 ns	slm	0.3	1.44	BPL	2,950		43.2
355	0.56 ns	slm	0.5	0.87	Bubble	330	2.62	47.3
355	0.56 ns	slm	0.5	0.87	BPL	1,050		133.4
355	0.56 ns	slm	0.8	0.54	Bubble	140	2.83	55.8
355	0.56 ns	slm	0.8	0.54	BPL	380		96.8
1064	1.02 ns	slm	0.8	1.62	Bubble = BPL	4,210	3.83	23.2
532	0.95 ns	slm	0.8	0.81	Bubble	840	3.30	49.5
532	0.95 ns	slm	0.8	0.81	BPL	1,200		48.5
355	0.93 ns	slm	0.8	0.54	Bubble	270	2.45	63.1
355	0.93 ns	slm	0.8	0.54	BPL	770		66.5
1064	11.2 ns	mlm	0.8	1.62	Bubble = BPL	25,000	1.59	2.7
1064	11.2 ns	slm	0.8	1.62	Bubble = BPL	80,800	5.15	24.9
532	8.8 ns	mlm	0.8	0.81	Bubble	2,000	0.65	1.1
532	8.8 ns	slm	0.8	0.81	Bubble	7,500	2.43	4.8
532	8.8 ns	slm	0.8	0.81	BPL	84,400		7.4
355	6.8 ns	mlm	0.8	0.54	Bubble	600	0.56	0.7
355	6.8 ns	slm	0.8	0.54	Bubble	1,400	1.31	2.1
355	6.8 ns	slm	0.8	0.54	BPL	42,400		9.1

Table 2 Thresholds for bubble formation and bright plasma luminescence (BPL), and threshold sharpness S for various pulse durations τ_L, wavelengths λ_L, and modes of laser operation. Modes are: ml = mode-locked (sech^2 pulse shape); slm = single-longitudinal-mode (Gaussian pulse shape), mlm = multi-longitudinal-mode (statistically varying temporal profile as shown in Fig. 5a). Irradiance thresholds I_{th} were calculated from the diffraction-limited diameter of the focal Airy pattern, d, and E_{th} using Eq. (5). When the BPL threshold is larger than the bubble threshold, only the energy threshold is given. Here, the plasma is significantly larger than the focal spot and irradiance values referring to the focal spot diameter would be erroneously high.

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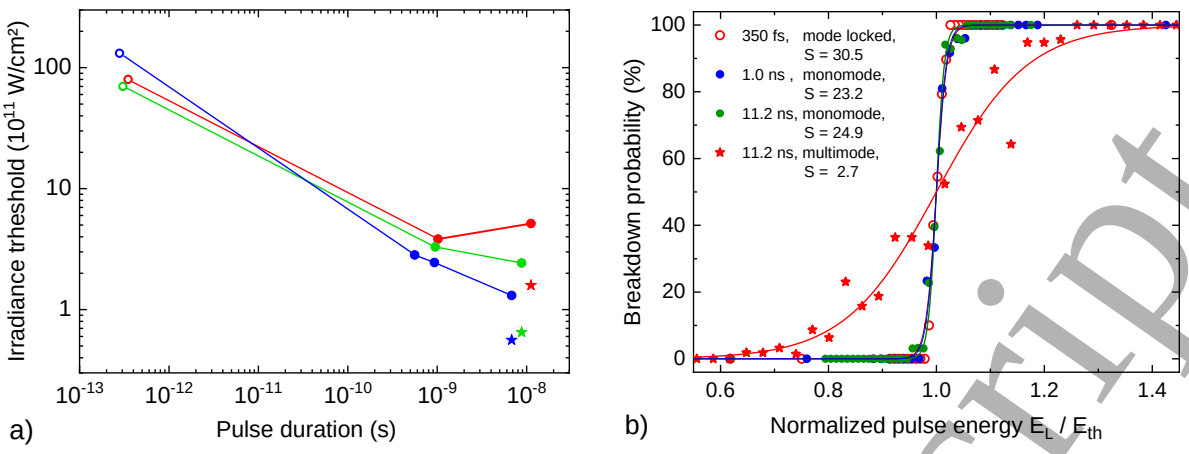

Figure 4 (a) Graphical illustration of the pulse duration and wavelength dependence of bubble irradiance thresholds based on the data in Table 2. The wavelengths are indicated by the colour of the symbols and the mode of laser operation by their shape: ○ mode-locked, ● single-longitudinal mode (slm), ★ multi-longitudinal mode (mlm). To guide the eye, straight lines connect data points for ml and slm operation. (b) Dependence of threshold behavior on the mode of laser operation. Probability curves for bubble formation in water by IR laser pulses focused at $NA = 0.8$ are shown for mode-locked fs pulses (350 fs, 1040 nm), slm ns pulses (1.0 ns and 11.2 ns, 1064 nm) and mlm ns pulses (11.2 ns, 1064 nm). Threshold sharpness S is high for the fs and slm ns laser pulses with smooth temporal profile but low for mlm ns pulses exhibiting irregular intensity spikes (Fig. 5a).

In Fig. 4(a), we see a pronounced decrease of the threshold irradiance for bubble formation with increasing pulse duration and a weak dependence on wavelength, both in agreement with previous work [36,37,108,114,115,125]. For mlm ns pulses exhibiting intensity spikes (see Fig. 5a), the threshold irradiance averaged over the spikes is lower than for temporally smooth slm pulses. For visible and IR wavelengths, slm irradiance threshold values change little between 1 ns and ≈10 ns, as predicted in [22].

Figure 4(b) shows that the threshold sharpness depends strongly on the mode of laser operation, which influences the reproducibility of the temporal pulse shape. Bubble thresholds are equally sharp ($S \approx 25$) for slm ns pulses with smooth temporal shape as for fs laser pulses. By contrast, for ns multimode pulses exhibiting statistically fluctuating intensity spikes, the threshold sharpness is reduced to $S = 2.7$. Generally, the threshold sharpness is 3-9 times higher for ns breakdown by slm pulses than for the corresponding mlm case – both for the bubble and the BPL thresholds. The threshold variations observed with mode-locked and slm laser pulses resemble the pulse-to-pulse energy fluctuations of the respective laser systems. Thus, with appropriate mode control, ns and fs optical breakdown thresholds are equally "deterministic".

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4.2. Energy dependence of cavitation bubble size

Figure 5 shows the dependence of maximum cavitation bubble radius on laser pulse energy for different pulse durations and wavelengths. Again, ns breakdown is highly irregular for multimode pulses [Figs. 5(a) and (b)] but well reproducible for slm pulses [Fig 5(c)].

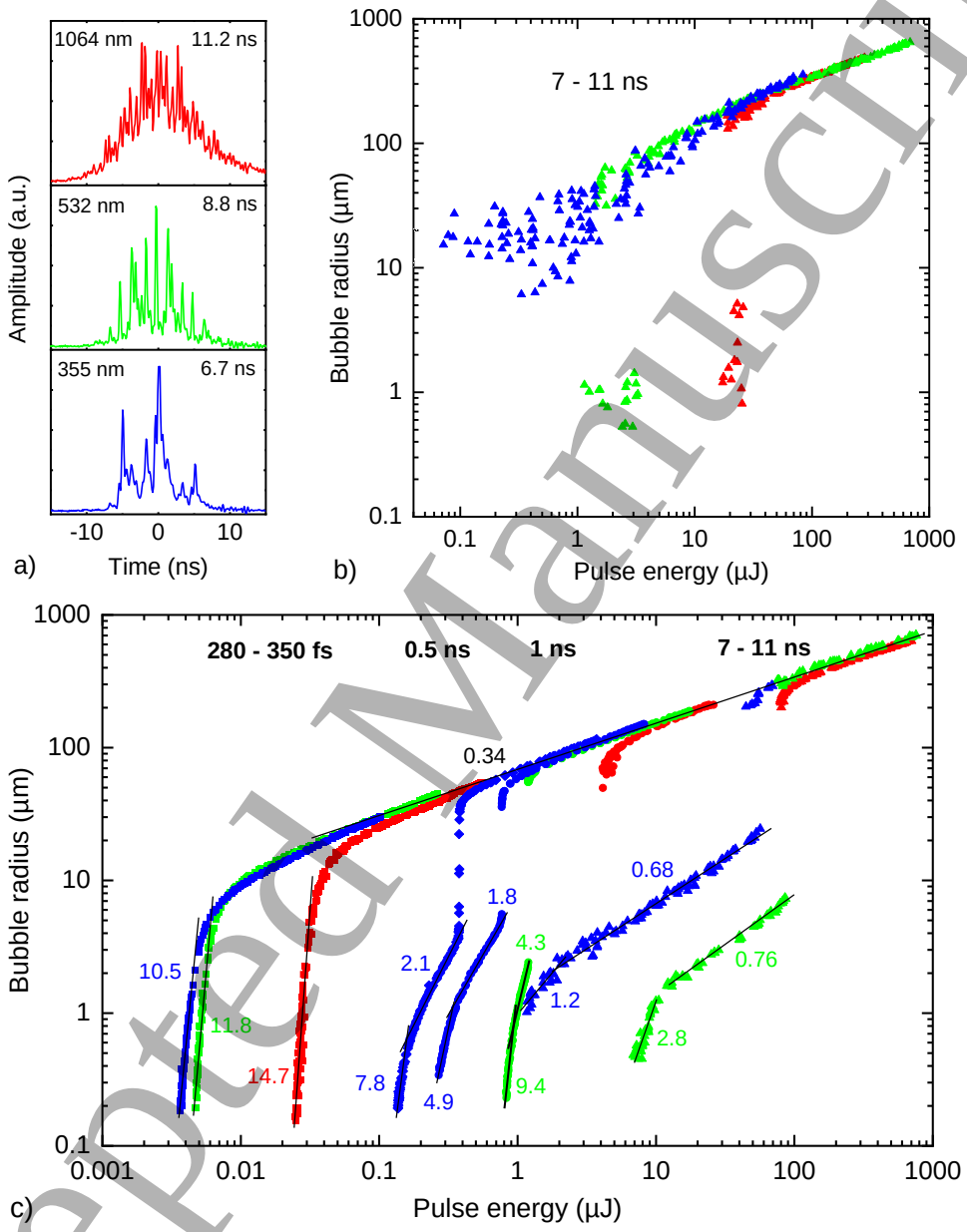

Figure 5 Maximum cavitation bubble radius as a function of laser pulse energy for different pulse durations, wavelengths, and laser modes; $NA = 0.8$. The colours denote the wavelengths: blue = 347 nm for fs pulses, and 355 nm for ns pulses; green = 520 nm and 532 nm, red = 1040 nm and 1064 nm. Examples for pulse shapes of longitudinally multimode laser pulses of 7-11 ns duration are given in (a), and the $R_{max}(E_L)$ data for these irradiation conditions are presented in (b). All ns data in (c) were obtained using longitudinally single mode (Gaussian) laser pulses. The fs data refer to mode-locked pulses of sech^2 pulse shape. In (c), the $R_{max}(E_L)$ curves are piecewise approximated by straight lines, with slopes indicated. The $R_{max}(E_L)$ data refer to breakdown scenario 1 for fs pulses, to scenario 2 for IR ns pulses, and to scenario 3 for UV and VIS pulses with duration ≥ 0.5 ns.

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